4 – October Focus Group – FG1

Interviewer

As with the previous focus groups we'll just kick off with a bit of an introduction because I don't know if you necessarily been in focus groups with the other participants here today. So, we'll just go through kind of the type one diabetes technology using, which smart watch you're using as part of the study, and this week kind of like a maybe a hack or a little trick or gadget you've got that really help you found helps with your type one diabetes management. So I go first. Yeah, as I've said previously, I'm on the t:slim and the Dexcom G6, which has kind of been really nice kind of getting to grips that in the last few months and the benefit that's seen. I use a Fitbit Sense just as part of the PhD, just to try and get a feel for things um and that's been quite interesting kind of changing. I find myself using it a lot less, that kind of shift over time. UM, But yeah, definitely still times it's kind of quite interesting to see what it says on there. And then I guess my hack little tip or trick or something I found that was really useful is just around kind of using exercise to help kind of supercharge insulin, I think I've heard people call it before. But if I have got high blood sugars to really kind of drop it down, you know, just going for a walk after I've given that dose of insulin. I really find helps drop that and kind of use of it. I think there are times it's maybe gone a bit too far, where I've kind of been trying to maintain those results and I end up kind of running on the spot in my flat which you know maybe is, yeah, a bit too far, but anyway. It's funny the things you do. So yeah, [P19] do you mind going next?

P19

Uh, yeah. So I'm I'm [P19]. So, the technology I'm currently using is the t:slim and the G6. And the smartwatch I'm using this my Apple Watch, series seven, I've had for a little while now. I don't really know if it counts as a as a hack or anything but one thing, one feature of my pump I use quite a lot is the um exercise feature and not just when I'm doing exercise, but I find it's just useful if I think I might be gonna go low, for if I'm working or whatever. I stick the exercise feature on and I've just found it to be really, really solid in kind of stopping hypos.

Interviewer

Yeah, I I think that is a really interesting point and and that's something I've definitely found as well. You kind of end up using it for other situations, well, I think. Like if. I think like drinking and overnight and kind of managing that drop, I've definitely found it useful there.

P19

And and driving in as well. I think if I if I could go on a long journey, I would just stick the exercise feature and I actually wish it could be called something other than exercise just. I don't know, just cause it feels weird putting it on when it's for exercise, but yeah it it's useful for other things.

Interviewer

Yeah, 100%. Uh, do you mind picking someone to go next?

P19

Yeah, [P16].

P16

Um. I'm on the Medtronic 780 pump with the Guardian 4 sensor and I use the Fitbit Luxe at the moment. And I think my kind of hack is kind of along the same lines of if my level's high, then I will like, go for a walk like [Interviewer] said. But also I find that for me, going to sleep actually helps drop my level, which is quite interesting.

Interviewer

Yeah, that is really interesting. Ohh yeah, I have to watch out for that. What kind of, how did you discover that link?

P16

Um. I think I noticed it when, in the evenings, my level would be high, and then I'd go to sleep and then my level would come down. And then also like in. So, like I think there was one time where I was on a car journey and my level was quite high at the start of it, and then I accidentally fell asleep and then I woke up at the end and my level was back in range.

Interviewer

No, that's great, you found, yeah, the little the useful way of managing it a bit. Uh, yeah, helping with that management process. Uh, do you mind picking someone to go next?

P16

Um [P02].

P02

So I'm on the t:slim and Dexcom G6. Um. I also use an Apple Watch, which I had before um doing the study and yeah, it's not being too different for me. But I'm actually almost finding the reverse like having. It's probably also partly to do with my work, but I which but I've also I found that I was wearing my watch a little bit more, recently, and now when I go to look at it and like go look at my wrist and I'm like, "oh, I've not got it on", or if I've not got it on, I'm looking for it. So, it's quite interesting. My hack, I don't. I don't really know. A lot of what you guys have already said like with the walking and the exercise feature, I do that a lot. But yeah, not just for exercise. But something

that I suppose I've picked up on or noticed quite a bit more recently is sort of when I have high blood sugars, like I have particular times of the day that I'm more likely to have a high blood sugar. And I've noticed that, if I have got one and it's not coming down then, I'm like, “OK, well, why” like cause I've done, and I should have done the insulin I've done. But I often have realised that caffeine um increases my, makes my insulin resistance worse. Um. So, if I've had tea or coffee, coffee in particular, I find quite bad. Um. It really like drives it up and then it will get stuck. Yeah, caffeine or just generally dehydration. So, not having enough water. So, like if I have only drunk that I've only had a cup of tea or coffee and haven't had a pint of water or something like that, then I mean, that's probably why and have a drink of water and about an hour later and that’ll come down nicely. So yeah, I've kind of paid a bit more attention to stuff like that for unexplained difficultness. Yeah, yeah, being a pain in the bum.

Interviewer

Yeah, that that's really interesting, you'd like to discover these things. I think it's so interesting these weird dynamics you find, particularly ones you're not warned about or told necessarily. Yeah. I I think I'm I guess, yeah. [P07] do you mind?

P07

Sorry forgot to unmute. I'm using the Omnipod 5 and the Dexcom G6 and I use an Apple Watch as well, which I did also have before the study. My hack, I don't know if it's a hack, but I started using Carbs and Cals again recently and I noticed now on the app they've got a lot of like chain restaurants and things and Domino's places like that which is quite useful because it wasn't always as easy to find the carbs for things like that. But. Yeah, that's probably my hack.

Interviewer

Nice. Yeah, that is a useful feature they’re beginning to add in and I guess it makes a lot of sense, there probably is a recording of that somewhere. Oh, that's pretty good and interesting that kind of this group here are all using closed loop systems, so that's probably probably something like to. Well, it may direct things slightly differently in the focus group today. The first thing. Yeah. So what we. I think the the first point of where I really want to start um, is just around this idea of kind of BG profiles and what I mean by that is kind of that change in blood glucose stuff as you see like the typical CGM graph like what your blood glucose have done over the course of the day or yeah, how that changes. And in comparison, what might you, you know what you aim for or what you might view as the ideal, with each other. So, I'm hoping to utilise the whiteboard feature to maybe help with this a little. I haven't yeah, use this with others. Is that kind of appeared in Zoom for everyone? Can they see?

P19

Yeah

Interviewer

Brilliant. And then if I, a little bit, but uh. Let's see if I kind of draw a. Is that appearing for everyone these different?

P19

Yeah

Interviewer

So yeah, if we use this kind of. What's this? This will be BG, uh in mmols uh, so the different levels and then you've got just time on the bottom. Um. So yeah, feel free to draw along this sorry, it's a bit kind of scrawled in there, but hopefully you kind of understand the basic of what the the graph is trying to show. So yeah, I I. Think just for the course of this, uh, some of it might involve kind of pointing out or drawing out what you're talking about, just kind of use the pencil feature. Draw a line, maybe adjust and say kind of different levels or something like that. If if you find it helpful, if not, don't worry about it. But. Yeah, I guess maybe to start off. It's useful to hear what you think is the optimal blood glucose profile or what you view as that. If anyone's happy to kind of jump as a starting point.

P02

I mean, I personally, I always remember or for me like my range. Like I'm happy if I'm somewhere between like 5 and 12 and relatively stable. That and I know that's like a much wider range than like the ideal like, but I've learned basically over time, the ideal is literally an ideal. It's kind of, I always find it's a bit of a fantasy because you're never going to have a straight line that's stuck at like 5.6 or whatever it is. Um. And yeah. And what is annoying for me is if like I've done my insulin and it goes up and then it gets stuck. Because that, that's particularly after lunchtime, that's a bad day and then end up being with the horrible crash. So, if I'm sort of relatively, what I call rolling hills rather than mountains and peaks and peaks and troughs, then between somewhere between 5 and 12 throughout the day then I'm happy. And to be honest, when I go below 5, I start to like, I don't find that terribly helpful for me because like I'm more at risk of going low and then I'm having to like preventatively treat that. So, if I'm going below 5 like it means I then have to think about “ohh if I want to spontaneously go out or drive or you know go for a walk, I'm having to take like like eat Jelly babies like you know”. And I'm like, “I don't really want to have to eat those if I don't have to”. So yeah, if I'm sort of a nice above, around 5, then I'm not worrying about going low, because yeah, the. Or putting the exercise feature on to prevent that. I'm sorry I didn't realise I haven't really drawn on the graph.

Interviewer

That's fine. It's more there as a tool if you want to use it. It doesn't. Don't feel you have to draw out, I think, yeah. You're explaining it well, it’s easy to understand. I like the rolling hills versus mountains analogy. What about other people? How does that compare to them? I think it's worth highlighting that you know, everyone's different, so don't feel like you know what you're aiming for might be

different. You shouldn't feel that just because someone else has something, yeah, you should be doing the same thing. But yeah, does anyone do the same? Agree with that or different?

P07

I think I agree with [P02]'s. For a while, I was really trying to achieve that flat line and obviously it's really not realistic. So, I think definitely try and be as flat as I can in between meals, after meals a slight peak, but hopefully not a massive peak. But again, it does vary on the time of day and the food. And then yeah, just trying not to dip. I think I find with this new pump, it's a bit easier. I used to kind of manually override and then it would drop down, especially if I saw a big peak on my CGM. So now I'm trying to trust this pump and just let it work it out itself.

Interviewer

Yeah. Anyone else wants to add to that?

P19

I I think when when I first got the the the t:slim, I had this bit of a habit of kind of chasing the numbers a little bit up and down. So, because because obviously the CGM date is like right there. So, whenever I was going up a little bit, I just like hammer in the insulin and just try and push it down. And it was doing two things, it was like stressing me out, and it was like, you know, making me worried more about diabetes than I should have been. And also, it was making me go have more extremes. So, I'd go from high to low to high to low and all this. And I think after a little while or after like about three months or so of using the t:slim, I just kind of, not really you know don't really take a laissez faire attitude, but more just kind of I stopped trying to get it perfectly flat and just accept it that it's going to go up and down and just take a bit more kind of a practical approach than than try and get a perfect kind of flat line.

Interviewer

Yeah, that's interesting. You were kind of like saying similar things there. I think, [P02], you mentioned it initially and I think the interesting point is that they're kind of being that, that kind of level you're initially told to aim for. I wonder if everyone's talked about like where they initially were told that or that information came from that side of things.

P02

Well, I think I think for me like, when I was in paediatrics like one, I had a very good team and a very good consultant, but it was often like. They see you more often in paediatrics as well and also you'd go in. I think because I was diagnosed when I was 13, um like, I went in and would would see them and it was, it's I was because I was personally managing it as well. Like it's it's different, I think it's different from when you're like really little like, you know, you have more parental input. But I guess for me like when I was diagnosed, I went straight in and it was like, “right, OK, this is now my

responsibility". And it's like. You know, they were always the paediatrician were fantastic but they always encouraging me to like, you know, they're looking at the HbA1c and they wanted me to have it down and it was kind of. I think yeah, because you you were going more, I was going more often as well. Like now I only go to clinic like once a year I get a HbA1c once, like once a year. Um. And like I think at the time, like if it was too a bit high like you climb, I kind of. I don't think, I think to say that they were disappointed in me is a bit extreme, but like they wanted you to have it down like lower. Um. And like I think I always just found that quite stressful and then it wasn't until I went into adults and um you know, and had a very difficult period like where I was constantly like, feeling like it wasn't good enough because I wasn't getting, like these perfect things. And like my, you know, my current consultant turned around and all the team turn around going, "you're doing really well. Like, you know, we're really happy with it. Like, it it doesn't have to be like. You know, ridiculously this line". And I think like life is just a lot like and I'm someone who's never, at least, you know going through a uni, going through school and that just life is not the same for me, like every day. So, how can you expect your diabetes to be the same every day at the same time? It's just not realistic. So yeah, I had changed my mindset over the course of, you know, the duration of my having type one and actually being like if it's between 5 and 12 then I'm you know I'm happy. And as long as it's not impeding me from doing anything that I want to do. Like you know if I you know I don't wanna be feeling unwell, um you know tired cause I've got a really high blood sugar and that. I kind of aim for that and kind of don't try not to compare myself to what other people are doing because, you know, everyone's life is so different. Everyone's biology is so different like what, you know, that diabetic person on Instagram with the perfect trace or whatever is doing is just not helpful to compare yourself. It's like how you feel. You know what I mean? So as long as, as long as you can do what you can do and it's you're those levels or whatever and not causing me problems, then I'm kind of happy with it if that makes sense. Turned into a bit of an essay that, sorry.

Interviewer

Something kind of that's been mentioned by most of you there is that is is this isn't necessarily a consistent thing, what you're aiming for. There are times when it changes as a result of different things. I was wondering maybe [P16] is there, are there any times in particular where you've noticed a change in what blood glucose you're aiming to try and maintain or range you're trying to maintain within?

P16

Um. So, I've just been away on holiday and we did a lot of walking. Um. And I think that I was probably aiming my level a little bit higher. Um. Because in my head I was worried that it would it would, if it were sat, say, at six, I was worried that it would be dropping. But actually. So that was what I was going to aim to do, like to have it like 8 or something, but um I found that most of the time, on the walks, my level was sitting about 5-6 ish. And I thought I'd be really anxious about that, but because I didn't have much active insulin, I was happier for my level to sit lower. And I think when my levels are lower, but I have a lot of active insulin, that's when I don't like it. But when I think it for me a lot of the time it does depend on how much active insulin I have on board. It it um changes how I feel about where my level is. Like if my level was. If I had like 2 units active and my level was 8, I'd be, I wouldn't feel anxious about that, but if I were 6 and had about two units active for my ratios, that would make me feel quite anxious.

Interviewer

Yeah, yeah, I guess that those are two really interesting points. Yeah, this activity impacting it. And then also the active insulin and that’s role within it. Does anyone else have any kind of times or factors that they find impact what they're aiming for?

P19

One one relatively recently or back back a few. Well, was it was quite a while ago now when I was doing my dissertation at uni, um and I was basically just said to myself as long as I'm not like like hypo and as long as I'm not like so high that I'm like ill, I don't really care what's happening between. Because I was like I just don't have the mental capacity to, like, do my usual kind of, you know, pretty strict control over the diabetes. So, I was like as long as I'm not a danger to myself, then for this next it was kind of the week before it was due in, I was like, you know, to hell with the targets kind of thing. You know, if I need to like cruise at like 12 or something, then I'm not, not a problem. Because I found that I was constrained so much, I just kept going hypo. Um. So, I basically have like reduce all my insulin and everything. But I was just I definitely get a lot less picky at certain times when I've got kind of just bigger things to worry about, to be honest. Because I think you know at the end of the day, as long as I'm not like ill or going to go to hospital I think there are, there are bigger things to worry about sometimes than getting those those perfect numbers.

Interviewer

Yeah. How do. What do people think about that? Is there anyone who finds the same or?

P02

Yeah, with you on that one, [P19].

P16

Yeah, me as well.

P07

I agree.

Interviewer

Yeah. No, it's it is interesting. Yeah, general life as well and the other kind of how busy you are and what that kind of plays into it and that role. Um. Yeah, yeah. I wonder if. Yeah, I wonder if people could maybe go into more detail on that. I think that's quite interesting yeah, concepts, how much you're doing in the rest of life impacts the management you're aiming for. Um. And maybe yeah,

detail, I don't know it, what what kind of you're looking to minimise or what kind of that ideal control is in these situations.

P07

Well, kind of like what [P19] said. Like I'm back to uni just now, so I find got a lot of back-to-back classes. So, I'm not really sitting looking at my numbers all day and I'm not really able to correct them because obviously I have permission to do that, but it can be quite awkward if you're in a small class taking out what looks like a phone, and if they don't know it's a bit awkward trying to explain it. So, um I think in that time, like everyone's said, as long as I'm not high or low for a long time, I'm not as worried. But I think having the continuous glucose monitor and the pump um that can adjust the rates, that's really helped because they have a bit more peace of mind that it's going to adjust itself and I don't need to do as much. So, I find that really useful, especially in labs it's quite hard to take out your pump and things, so doing it itself is really useful.

Interviewer

Yeah. No, no, definitely. And I, I think building on that. Yeah, I I think it. It's quite interesting that we do have a range of kind of closed loop systems people are using or hybrid closed loop in terms of t:slim, Omnipod 5 and the Medtronic 780. The kind of activity feature in the t:slim has been brought up. I guess, uh, two things, one maybe in the other systems or maybe in the t:slim as well. Are there other features to use to try and adjust that kind of target and how you manage around that? And the second, is there anything more you'd like to see or anything changed within it to better, yeah, help you to adjust what you're aiming for?

P07

Um. So, I'm using the Omnipods 5 and there's also an activity feature on that so you don't control like if it's putting more or less in and it's just aiming for something like 8.6. I can't remember exactly. I think I did mention this last time, but I think if there was an option to kind of choose which activity you did, it would be useful for it to learn what your rates normally change when you do that activity, because they've found if I go swimming I probably drop quite a lot, but then if I'm going to the gym it might rise. So, it would be useful if you could sort of choose which one you did in advance, so it new “do I need to turn off the insulin or adjust that to put less in, or do I need to actually put more insulin in?” I think that would be useful.

P02

Yeah, that that sounds like something I would find very useful as well, what different activities impact what my blood sugar does. The other thing I was going to add is like for me, the one thing that I probably have consistently struggled with is like before you do any kind of activity or whilst you're doing an activity like and you have a meal. Like, for example like if you're doing a long like a hike or, you know, you're out just generally very busy like that, how much insulin you bolus, like, how much? I have struggled with that for so long because, and this is like where my human error gets, messes me up. It’s because I think ohh you know I do this amount and sometimes it's too much, still too much. I

wouldn't. I don't reduce it enough. And then, you know, like, half an hour later, like I've resumed my activity like my hike or whatever. And I've gone low and that's really frustrating cause now I'm not only full, but I'm also low as well, so I feel like disgusting. Or you know, you you haven't, you reduce your bonus for your meal too much and then you feel high and you're continuing to go for a wee like and it's just or you feel dreadful and you know. If it was some kind of yeah, adjust. Like if the pump could take some of that human error out of it and be a little bit more, I suppose. Yeah, if you could. If there's a way to consistently get like an adjustment to bonuses for exercise, that would be nice.

P16

With my pump, so I have the Medtronic 780 and I use the Guardian 4 sensor. It has the exercise mode which it aims to get your blood sugar to 8 point something I think but. So, it is very helpful, but when it's in that mode, it then doesn't do the auto corrections. Even if. So, it'll just give you a bit more base if you go higher, but it won't give you any like little corrections and I think it would probably be more useful if it did give corrections. Obviously, not as much as it would if it wasn't in there, in that mode. Because if I'm not doing a lot of exercise and I put that mode on sometimes I just go really high because it's not giving enough, if that makes sense.

Interviewer

Yeah, definitely. I I think it's interesting these points you're making, yeah. And at least you got really good ideas to kind of expand the, yeah, kind of the reach of these exercise functionality. Yeah, the exercise feature. Sorry. Yeah, both into one of those. Yeah. I guess like maybe focusing it on on the yeah, the meal side and kind of those carbohydrates are making adjustments there. Does anyone else have suggestions for ways that could be adjusted to kind of help with meals at those times?

P02

Just to expand on or to add to what to what I said earlier as well. I don't know if like it does this or not but like on the t:slim. But the one thing again like with this whole like pre bolus thing or like you eat a meal, you go out for doing activity do exercise or you know. Because obviously the insulin window, like is active for three to four hours depending on what insulin you have like. You know, you might have your lunch and then like an hour and a half or two hours later, go out and do an exercise. But you put the exercise mode on, but I don't know if it if it like, really takes into account what insulin you've got on board. Because, like, for me, I might have started. You know, I might have had my lunch or whatever two hours ago, then there might still be a bit of it. And I might be like nine or eight or something and that would, that's great, like, but then you start the walk and that remaining insulin then drives you low. But like the exercise mode, won't. Basically like if it was me doing it manually, I would just turn the whole thing off. Like but you put the exercise mode on and it that doesn't turn it off like it leaves it running for a bit and then you go low because it hasn't turned it off and you're like, really, I could have just done this, like, prevented this whole low. Like, yeah, like, so I feel like you know the exercise mode is helpful, don't get me wrong, but you still, for me, like, requires so much like cognitive effort to like think on “what's on board? What have I eaten? What am I doing?” Like and it's still. Yeah, it's definitely not foolproof.

Interviewer

Yeah, I think that that, that sounds like a really an interesting point. Yeah, based on the insulin you have on boards changing. But I guess it even comes back to kind of like the point [P16] was making earlier and that kind of insulin on board point, maybe it acts more or less aggressively based on that. Um. Yeah, that's really, really interesting. And I guess it's something that leads maybe leads from that is obviously quite commonly that's based of kind of information you've entered in terms of the carbohydrate you've had or the activity you've done. I I was wondering how, how accurate you would say that information is that you're inputting to the closed-loop?

P07

I would say if I know I'm going to exercise, I tend to maybe put less carbs in for the meal before. Um. Like for example on a Sunday I do swimming in the morning and I always eat before, but I'm worried if I put a lot then I'm gonna drop very quickly into the session. So, I think definitely [P02]'s points of having a way to get the system to recognise you have just eaten and adjusting it accordingly would be really useful. Because like [P02] said as well, when I have my old pump, I used to just turn it off. Obviously now you can't do that, so we could have that as an option.

Interviewer

Yeah. What about others here? Yeah, does that? Is that similar to them or yeah or are there any other times they think maybe they they adjust what they put in based on kind of like trying to get the closed-loop to do a particular thing?

P16

Sometimes I'll notice that the closed the closed-loop system isn't giving me enough basal before a meal because I've noticed that when I put my put a bunch of insulin in for a meal, often it will stop giving me background insulin. But I know that if I'm gonna be in a meal that I need that background insulin before, because otherwise I'll just shoot up. So, I'll often, if I know that's going to happen, give extra insulin to combat that.

Interviewer

Yeah, I I think that's that's an interesting point. Yeah, and interesting like the differences and. Yeah, that might be, yeah kind of like a personalised thing to you. Is there, what would you want? I guess could you think of a way that you'd be able to kind of explain that to a closed-loop or is there something you'd like to be able to adjust on a closed-loop to kind of allow for that to be more?

P16

Um. I don't really know, I think. Yeah, I don't really know.

Interviewer

Yeah, what about anyone else? Does anyone else have suggestions? Maybe ways that could be done or similar features that could be added?

P07

Don't know if there could maybe it's kind of going back to combo bolus, but you like with the exercise feature, you could have an option that's like I'm going to the a take a way, or a fattier meal or something, and it can maybe just cause you know you're going to rise quite a lot. So, have an option to put that on if you think your meal is gonna make you rise.

Interviewer

Yeah, I think that that sounds like an interesting point. A way of maybe managing that situation.

P02

Just to add to that point? Sorry.

Interviewer

No, no, go ahead.

P02

Does anyone else find that pizza, absolutely destroys their profile like after you eat? Just, like I love pizza, but now it's so problematic I only have it on such a rare occasion because. And just going back to like what we're talking about earlier, like [Interviewer] is like about when it what your range changes at different times. Like after I've had pizza, if I'm like, you know, not like 20 or something or, you know, if I have like a nice profile after pizza, I'm chuffed. Like, you know, so like my range, just so you know, just the fatty foods and stuff like. Yeah, if you can find a way to input like a pizza profile, that would be nice. It's like. Yeah. Sorry, just what [P07] totally all of a sudden just reminded me of that.

Interviewer

No, I think it's interesting pizza gets brought up. I mean has been brought up like several times with various people in this study in the previous one I've done. It's funny that like that one food source has like such a reaction from people about. Yeah, you can kind of comments on that it is quite fatty food and that's slowing it down and also being very carbohydrate rich. I think it's quite an interesting area that. I mean, in some ways of potentially dealing with that is kind of giving it extra information. So, either and whether that's the form of it is literally a pizza setting saying this is pizza or whether you then have to start to input fat values as well as carbohydrates. Um. I just wonder how people felt

about that? Almost the extra effort that that is from having to give more information compared to, yeah, the potential improvement in diabetes management as a result.

P07

I don't think I would personally mind too much, because if you're waking up the carbs anyway, things like the fat is probably on the same like place like on the packet or on uh carbs in cals. So, if it would help to stop those spikes and the drops after I I don't think I would mind at all.

P02

Ohh you could even have like in the same way that you know, like the Dexcom and stuff like talks the pump, you can always you could all these like apps like you know like the carbs and cals that. I use is slightly different like one that you know you can look up exactly the food content like the sorry the macros in the food. And if that was, then communicate to the pump, you wouldn't even have to import it and it would be quite accurate. You know that would probably be a lot more, give a lot more information than just carbs. I think, yeah.

Interviewer

Yeah. What, what do other people think about that idea? Kind of almost integrating the carbs and cals app into a closed loop so kind of look up the food and put in the grammes rather than kind of searching for the carbohydrate value and then putting that into the closed-loop.

P07

I think that sounds like a really good idea, especially if you're already using the app. And it kind of actually takes a step out for you, so I I like that idea.

Interviewer

What about you, [P19]? Is that something that you think you'd be interested in using or?

P19

Yeah, I think I think, I think if it was easy to use and yeah. It it's a weird little thing, but it comparing the physical form of the t:slim pump, the Medtronic pump I so much prefer using my t:slim pump because it's a touch screen. I used to find the Medtronic pump mind numbingly tedious, having to scroll through all the numbers to get to the the number that you want and I think that's along the similar thing. Anything that's easy to use I would use. I think if it's. If it's a if it takes too much effort on just gonna bin it off because it's kind of cost benefit analysis really. You know, can I, is this really worth the extra effort? Is it gonna make a noticeable difference? If it’s going to make a noticable difference, then I might put a bit more effort into it. But, it's only going to make a little bit of extra

difference then I want it to be as easy as possible. And I think like having a connected app probably would be be pretty easy to be honest.

Interviewer

Yeah, I I think this, it is a really important point that the value gained from the effort put in. Um. How do you all think you would balance that out? Is there a particular level you'd be happy with or how would you even find that level just through testing or other methods?

P02

I think if it's quick and reliable, like something that isn't really time consuming. Like, you know, [P19] was saying, scrolling through numbers or like, stuff like that. And something that's reliable like it will always. You know you haven't got disconnect or reconnect it to make sure that you know or try and problem solve or like troubleshoot the issue that has gone wrong. Like anything that doesn't take up time because diabetes is so time consuming anyway and it like. I guess from my personal perspective I want I want my technology to like subtract like their time and the stress that goes associated with diabetes. I don't want it to add to it. Because it's almost like you know, I want my more of more of my time and life back, not more of my time and life invested into it, if that makes sense. So like. Yeah, so, so so I say quick and reliable like, then you know early. I think probably the most important things.

Interviewer

Yeah, I I think. Yeah, I think I'd like to maybe go into that, that more. Is this idea of yeah, the value you get back from it and yeah the effort you put in. I just wonder if. Do you think that changes or like how do you think that changes depending on how good your management is already? Say you've had had a very good management already, would that impacts kind of like your willingness to try out new features or do you kind of reach a point of acceptance of it and then you're kind of happy and stick with that?

P07

Uhm, I think for me if it would actually make a difference like. For example, if you weren't going to go right up to 20 or unrecognisable high number when you have a takeaway, then I would be willing to try it. Um. Just because even if you have good control, moments like that do make you feel pretty rubbish and and I find I get pretty bad night sleep if I'm up at that range. I've had a takeaway for dinner, so if it did make a difference, I wouldn't mind doing it.

P02

Yeah, that is like the primary reason why I so like so rarely have pizza now because like yeah, if you eat it in the evening it like you sleep like absolute trash and then you wake up feeling absolutely horrendous. It's like for me, like as much as I love pizza like it's not worth it. And like, that's kind of sad, but it's kind of the reality. But I think just, you know, going back to your like question

[Interviewer], I think it's probably quite personal or very personal. Like how you know people's, people have different thresholds of how much time or effort they're willing to, you know, spend on their diabetes. Like, you know, some people will, you know, spend like their whole. What, what might look like to someone else like they might spend so much time doing it like and, you know, let diabetes almost direct their life rather than their life directing the diabetes. But it's it's very personal, I think. But I think it's what I think it's just having the features. I think there's a lot of people who are, like I know I speak for myself here, that are still looking for improvements in technology to help them lead the life that they want to lead without such, without the difficulty or the extra things you know, consuming their time. And and I guess that's from my perspective it's like, if these new things as they develop are going to help me like or, they might help me, I'd be definitely willing to try it. If it doesn't help, then you drop it like. But you know, if someone's already, if someone's life is stable or they don't want to change it, which you might see in in some people like maybe older people who have had diabetes, like their whole lives or grew up a long time without it, and then, you know, they're very happy they don't need to change it. Then why would they? You know, I mean, it's I think it's just in increasing options for people. Because like I say, like everyone's lives are so different and for people who want, don't want their options to be or their lives or what they're doing to be restricted by the diabetes. Like more of those options need to be there for them if that makes sense. That's how I feel anyway, I said. I feel like I often speak like for everybody. I don't mean to. I'm. These are my opinion. So yeah.

Interviewer

Yeah, I I think that's. No, it it is a good point and well with many many things and type one diabetes it is so individualised and personal to each person. Um. Yeah. And I think even people have been talking about in this one and that changing over time as well. Talk about going through stages of life where there are different kind of aims or targets and the effort you know you have time or you have bandwidth to put in. Um. Yeah, thinking about this extra phone systems that kind of asked for that more information how how would you feel about being prompted for that information? Whether it's through a notification, maybe occasionally asking for that maybe the fat content of a meal or more details on what exercise has been done.

P07

I think for me it would depend on how often it was asking you. Like if it was constantly asking you, that might get a bit wearing and it might be quite hard to always be able to do it if you're busy or something. But if it was, say like after a meal, that wouldn't bother me too much because you're already having to put something in anyway. Um. The only thing I can think now is might be like these apps, they don't always have everything on it, so I feel like if it was to be used they'd need more information. And because things like carbs, I think now with experience you can I know you shouldn't, but you can sort of estimate from your experience. But with that I wouldn't really know where to start estimating fat, so there was more information I think I wouldn't mind being probed if I could easily find. What it wanted me to put in.

Interviewer

How about from an explainability point of view, would you want the system to be telling you why it's asking for that or what information? Would you want or would you prefer? Yeah. What, what information about why it's telling you or asking for this information would you want?

P19

I think I'd want to know ultimately what it's gonna do with that information. Um. Because it's a little bit like you know obviously when you first go onto the closed-loop system and everything, or at least my pump rep explained a little bit what the algorithm. So, what it does do, what it doesn't do. I think ultimately is going to be doing that because if it's going to be in any way feeding into diabetes treatment decisions, either via myself or via the pump, you know closed-loop system. I need to know exactly how it's going to affect. And also what you know for it's if it's asking for things like fat content, OK, what happens if you then don't put the fat content of a meal in? Is that gonna negatively affect it or is it kind of a neutral thing where it doesn't matter if you do or you don't, but it's a it's a bonus if you do. But I think it, that will be more like when you start up a new system with that feature. I think as long as they kind of they communicate the limits then that would be that would be what what would want I think.

Interviewer

Yeah, definitely. And how would you want that explanation presented to you, do you think?

P19

Um. I'm not sure. I mean I I quite. When I started the t:slim, I quite liked having a chat with the with the tandem pump rep and the pump kind of person from the diabetes team. I had good ones. I quite liked having it explained to me in person just because I feel like if you have explained to you by someone who knows what they're talking about. I certainly trust them more than maybe some material that was perhaps put together by some corporate team that don't actually know what they're talking about, but it kind of pushing the marketing features. I don't really care about the marketing, they don't really care about the features from a marketing perspective. I wanna know the the features from like a medical perspective. And I think for me it's easier to trust a person than it is to trust like a leaflet or an online kind of video or something. So may maybe a video potentially, but I think, yeah. Ultimately, I'd have to trust that they they kind of knew what I'm talking about, whatever form of information it is.

Interviewer

Yeah. How, how do other people feel about? That do they think like. Do you think you were given adequate information when you were moving on to the closed-loop you're on about how it worked and? Or to at least to a level that you wanted it.

P07

Um. We had like a call kind of like this. It was on teams though I think. I think it would have been useful if it was in person because it was quite hard to ask questions in uh, when there was a few of us. Because she was kind of explaining it like we were all on Dash, but I was on the version before Dash. And I kept trying to jump in and say, but I didn't have the chance. So, I was just trying to like, catch all what she was doing. Um. So, it probably was my fault for not getting her attention. It was quite difficult on a call, but if it was in person, it would have been a lot easier for me to kind of stop her and just ask her to re explain some of the things. But they do a lot of the training courses before you attend now, which is quite useful to just get a bit of an idea. And there's the booklet. Um. But sometimes the booklets are really big and I just don't know where to start with reading that.

P02

When when I moved on to the t:slim I also was online. Um. It was OK. Um. I also kind of share the sentiment of kind of it may have been better in person, but it worked. But the reason I kind of wanted it to be in person is I've gone from um yeah, no, the old Omnipod to having to use cannulas and I hadn't used cannulas before. So, that was slightly interesting., learning to put a cannula in over the camera. But, I think that's the thing about diabetics, we're relatively adaptable, so just kind of get on with it. Um. But I think, yeah, it was OK and it was fine in the end and they did have lots of videos online, which was OK. But yeah, I feel like some of those things, like technical things. Like information over online maybe OK but like when you actually have to do something a bit fiddly. I mean, we spent like a long time like she was trying to show her stuff over camera and it was it was OK. But like, I don't know, but. I just want to say like for you [P07] it's not your your fault that you know that the rep should have asked who was on what systems. I feel like don't blame yourself for for that. Like that's their responsibility as the as the rep to us, like the people there, you know. And I feel like that's kind of you know something that you know that as patients we shouldn't feel like we're the burden, you know, I mean like. You know they're there to you know. They're working for the companies who's getting the money out of the NHS for using these pumps, so they should be providing the adequate service.

Interviewer

Yeah, it it seems I guess a lot of these comments are around kind of the the process of using the closed-loop system. I I wonder how much interest people had into maybe kind of like the inner workings or the algorithms go into it and that decision making process. Was that something that people wanted information on or how much information did they want on that?

P02

To be honest, ohh sorry you go [P19].

P19

Ohh. OK, thanks. Uh, yeah, I mean I I. I certainly asked them very much some of the tech, some fairly technical questions, because I wanted to understand exactly what the what the algorithm was going be doing, what the limits were. The thing that surprised me was how inflexible it all is. As in, you

can't really change a lot of the values apart from your basic, you know, carb ratios and and basal whatnot. But I kind of I want to know exactly what, what I could expect and. They were generally quite good. They gave me some of the the, the, the booklets and stuff which had some of the actual physical values. Now I can't remember any of the exact values, but kind of when I was getting started it was quite helpful to to know what what I could kind of expect to how you know how? What kind of degree of correction I can expect it to be doing in the background um or what kind of levels it's going to be you know triggering certain alerts and everything.

Interviewer

[P02], was there something you wanted to say on that as well?

P02

No that's so good.

Interviewer

Um. Yeah. No, I I think it is interesting and that that initial. I I was wondering, [P19], if is is that something you wanted? Is that like an ongoing desire to like still know what it's doing more accurately or technically. Or was it just kind of at the beginning?

P19

I think it was more at the beginning um to kind of get me used to the the system, having been on the the Medtronic with the Dexcoms, the two things not talk together at all. It was more kind of a peace of mind thing. I was always very open to the technology, um I really pushed hard to get onto the system, so I absolutely wanted it. But for me it was, it was just helping that transition from me doing everything to a digital technological thing, having some degree of control over the diabetes. Um. Whereas now, I mean obviously if they ever had, I don't know whether they ever would update the algorithm and if that's what they do. But if they ever did, obviously I'd want to know what they've done. But in terms of an ongoing thing, not really, cause I know how it works now and I kind of I know how it works with my diabetes and myself. Um. So yeah, it was mainly just in that kind of initial startup.

Interviewer

Uh, what about you, [P16]? When you were moving on to the closed-loop what was maybe like the split of interest in just using it compared to how it was kind of working in reality, or the more technical side, maybe?

P16

So when I went on to the closed-loop, I'd already been on Medtronic pumps since I went on to pumps and I've been using the sensors alongside it. So it wasn't like a huge change, it was just cause I'd already had like the suspend before low and stuff. So, it was just the algorithm and the closed-loop thing that was different. Um. And I think I was given information about it. And I think what I was what I know about how it works is enough. I think for me there is such thing as having too much information and knowing too much about it, because I think that would just make me over think about it.

Interviewer

Yeah, I I think that's a really interesting idea. Um. I wonder if anyone else has any, yeah, views on that. Do they feel the same or differently at all.

P07

Yeah. When I first started. So, kind of like what [P19] said, I really had to push to get this pump. Like it must have been around a year, it was really asking and asking. Um. And I kind of use YouTube which I know cannot always be accurate. But obviously the Americans had the pump first, so I was kind of looking at reviews that people would put up and like some of the diabetic channels about what they thought of the pump. So, I think that kind of help me learn it like the basics about it and make me feel like I could trust it because you know, it had been out for a while there and it was going well. So, we did get some training when we did the call but like [P16] said as well, I I feel like I know the basics, but I don't feel like I personally need to know a lot of detail. We've had it for a while now and I still trust it. So yeah. I think that's just my opinion so.

Interviewer

Yeah. Yeah, I think that that that's really interesting idea. I think almost. Yeah. It's like particularly like the value in this sort of like focus groups and engagement with people is there is that variation and it's important. UM. Yeah, those opinions are reflected. Yeah. This maybe going back a little bit back towards more of the profiles. Yeah, the ideal BG profile. I think a few people were mentioning. Um kind of. This uh, yeah, the concept of like this ideal they're aiming for and kind of almost. Yeah, there are a lot of things set around particular numbers and I just wondered how people felt about kind of those core, these kind of like almost the. The kind of like trying to just capture diabetes management in a few numbers. Whether that's HbA1c or time in range and and there were kind of a few comments around, you know that that being quite difficult at times and there kind of other things to capture. I wondered if anyone might just kind of like expanding on that.

P19

I think like when when I after a little while have been on t:slim, like my pump, you know, team whatever they were like, "OK, you need to be". You know, before before I was on the pump, was on the t:slim I was like, you know, just over 50% time range, whatever. And they were like, OK, so we need to be aimed for like 65 and then next time they're like "OK try and get 70" and then and then. Now they're like, "OK, so we do over 80" and I am and that's all good. But one thing I said to them,

this is like I started when I was in uni and started started t:slim when I was in uni and I was like I said to them like. It's all a balancing act really. It's like, OK, I could get 80% time in range, but I could also be like failing all my uni classes because I'm devoting like so much time and energy into like the diabetes. So, I said to them is “I've gotta achieve this balance between, like my mental health and my outside life and my diabetes”. Obviously, I want to have good diabetes control, that's a given. And so I think with that, I then took a a bit more for like a bit more of a pragmatic approach to it. And I think again it was even these people who were like medical professionals were still they say they're trying to capture the whole diabetes system with just a couple of numbers. And I think what I was trying to explain to them is there's ways of measuring success which aren't necessarily the time in range. I mean that was the big one that they were hammering, not even particularly HbA1c, is the time. It was all about time in range, that's all they really cared about. But yeah, I kind of had to just say to them, it's gonna take me a little while to get into like those high time in ranges but as long as I'm doing well in the rest of my life I consider that a win.

Interviewer

Yeah. Has anyone else had similar or different experience with clinicians or other areas?

P02

And I've been very lucky, like my team have, in like that, my adult team have. When I moved. Well, I've been, I've been in. I was in the adult team when I was in uni and they were very, very supportive. I mean I had one. Basically where I am now and the team I have now are really nice and they've always been incredibly sort of, I suppose conscious of like peoples, like patients, mental health when they've come in. And that's, you know, when I've sort of said “ohh, you know, go through some rough patches” and they’ve said “you know you're doing really well” like you know as long as. They've never sort of reprimanded me if I'm having a bad time or anything like that and said “ohh you know you should be doing this or you should be doing that”. Like they've never said that and but I have heard like some people's teams being like saying to them, “oh, if you're not this, then we're gonna take away your equipment” or this kind of thing. And that just like, really makes me so mad because I'm like to hear, like, that people would patients are treated like that. Because, you know, not being funny but, you know, yes, our consultants, nurses, they are they know the medical side of diabetes, but they haven't. Most of them, haven't actually got it. Do you know what I mean? And it's sort of like, you know. Yeah, they want to look after us, but there's more to. There's more to life than diabetes, and that's kind of you know what, something I've had to learn. Like, you know, it's it's hard because, especially if you're someone who is a bit of a perfectionist and wants to wants to look after your health. I think when you're, at least from how I felt having diabetes makes you value your health a lot more. You know, it's made me value my health a lot more because of. Not just, not just because of the diabetes, but like the other associated problems that you end up, you know, coming along with it. Um. But yeah no I think. And and that's why like you know, I think it's really important that a lot of teams, like if they're not doing it that they should um have like onboard psychologist. Or like take a kind of, have that more holistic approach of yes is the diabetes is is good, that's great but, like how's the person actually doing. You know what I mean because a lot of the time if someone's struggling, you know, mentally then, of course, their diabetes isn't going to be great because they haven't got the capacity to look after it. Do you know what I mean? So, I mean, luckily I've never. My team have always been very good and and that I know that that facility is there if I need it. But it

does make me angry and really sort of sad that it's not the case for everyone. Yeah, and yeah, post code lottery, isn't it just where you live, the type of people working on the team.

Interviewer

I I think these are, yeah, really interesting points you're all making. Um. I guess going into the yeah, these ideas or maybe [P19] you mentioned around these kind of like different ways of measuring success. Um. Yeah. Did anyone have ideas on how that might be or what they would like to see being mentioned as well as kind of the timing range, I guess that was mentioned?

P19

I think part of it could be, I don't know whether you could really measure this, but it. Like, so when I was at uni like obviously I have to be a bit more careful with a lot of things but I I had a pretty normal ugly experience, to be honest. Or I had the uni experience I wanted to. And I think for me that that showed a good thing because the diabetes wasn't an overwhelming hinderance. Obviously, you know it had an influence and whatnot, did have some effect, but generally it didn't. Um. And I think it would just be nice if they kind of you know as well as the raw data they kind of consider the impact it's having on your entire life. But I have no idea how they would measure that to be honest.

P02

I mean just the. I didn't hear the whole of what you said, [P19], but I got the gist of it. I think my connection is probably rubbish. But um like just to build on that, like I, I agree. Like I think you know targets and time in range and stuff is helpful from a medical perspective. But like holistically like we are people who have got this condition at the moment, for the rest of our lives like. Me. You know, I might, but I have, like, a background in, like, neuroscience and mental health, so I I'm quite passionate about people's mental health. But like for me, you know, a patient who's got a chronic condition and, you know, not just diabetes, but like, I suppose, other conditions as well. You know, they've got to live with it like and actually it's. You know like a measure of success is really, is that person healthy? Like are they healthy and are they doing, are they happy? Are you know, are they doing what they want to do? Like if you know, if they're not doing what they want to do and they're unhealthy, well then that's a double negative. If they're. If they're doing what they want to do, but they're very unhealthy, then then OK, there's some problems. But if they if they're healthy, but they can't do what they want to do, then that's also equally as bad because they're not living so or they're not, you know, their mental health is gonna struggle. They're not doing what they want to do, really get to live one life and our poor buggers that we are, we have to live with all this extra crap. Like you know you shouldn't. You know, we we should in. How I feel is that we shouldn't have to be impeded by something you know, compared to to, like someone who hasn't got diabetes. We should be able to do the same things like or have the same opportunities. Um. So, for me, like, if I was a clinician like and a patient was coming to me like for me, a measure of success would be or is is my patient happy? Are they doing what they want to do? Is are they being supported to do what they want to do and are is their health allowing them to do what they want to do? Or, you know, is are they're not gonna, are they gonna run into serious problems like in the future cause they're health is even you know taking a toll or whatever. So, I know that that would that would be my, you know, measure of success. And and I guess that will look different for every single person that walks through the door

because. And I think that is again like they're at the boundaries between like modern medicine and kind of holistic, personalised medicine is, like you know, under under, like medical care is under a lot of stress to kind of meet targets and users like, you know, statistics and things to measure whether someone's normal or healthy. But actually, like, it's going to look different on every single person. It's difficult. But yeah. Hopefully that's answered the question. Or in my opinion anyway.

Interviewer

Yeah, definitely, definitely. Um. Yeah, yeah, yeah, moving on, kind of another area I wanted to kind of get yeah, everyone's thoughts on is just around kind of like privacy. Privacy of your data. I feel kind of the the trend in technology has kind of increased the amount of data and information available on the condition um and also the amount of information you input to it. How do people feel about the fact that this kind of quantity of data was growing?

P07

Um. I think it's a good thing because I'm hoping that means there'll be more opportunity for development of the technologies and even like in terms of the actual condition cures and things like that. If there's more data, there's more for people to work with to come up with solutions and new ideas.

Interviewer

Uh, yeah, what about anyone else, do they have thoughts? Uhm, I guess maybe from that then, who who do you feel comfortable sharing that data with?

P02

I mean, what are we talking about like blood sugar data, insulin data or like like. I suppose it. Like for me like my blood sugar readings like me and my blood glucose readings., how much insulin I'm having like, I don't really care. Like it's like I'm like a data producing machine. Like, you know, I'm numbers galore. Like, it doesn't really. I think as long as like it's not, you know somebody can't like use my data and trace back to me like. I think it's about what what the intent is that of the person accessing that data. Like if it's somebody we intending to use it to for research, then by all means like you know, I don't really care but and and well it's great like use the data. I'm not gonna use my numbers from the past like however long. Um. But I think it's the extent where you probably don't don't want someone to be able to like access that and trace you back to where you live and like or you don't want it to affect like the care that you receive, which is important. But in terms of like, yeah, people accessing like, which is why I was more than happy, like to take part in this study. It's like if it's going to be used for something useful and to help improve things or, you know, go into future research, then, you know, by all that's fantastic. Because, you know, in the era of information overload, like my data can be used for something useful then that's that's great.

P07

Yeah, I agree with that completely. That's what I would have said as well.

Interviewer

I guess maybe on the other side then is there, does anyone have any concerns at all related to it? I guess, yeah [P02] you were saying a bit about the intent, but kind of beyond that or more specifically.

P02

I guess my only concern would be I wouldn't want people or like clinicians to use the data to like, revoke people's technology. Like, you know, I was saying earlier like some places like offer, you know, like pumps or equipment on conditional basis like “ohh you know if you don't get this or take it away” kind of thing or that at least that's what I've heard of other people, some other people’s experiences. Hopefully it's not everyone but. I mean, but the thing is like when you're on a pump and stuff anyway, that same data is going to be available to clinicians, whether. But it's not. It's always going to be there because they need it, they need it to monitor your conditions. So, that even though we're producing lots of data that more data like. I don't think. In terms of like from a clinicians perspective, I don't think much has changed or like they might be able to use it to help improve your care. I don't know if that's not really made a lot of sense but yeah, I think. Yeah, I'm not really sure where I'm going with this one.

Interviewer

I think that definitely makes sense to me. I understand what you're saying that totally. Um. Yeah, and I and I guess part of what's been mentioned there, there have been several kind of different points that that, you know who that data sharing’s with. One is research and I guess the example we have is this case. Two was with clinicians and then you gets this the the more personal level of almost remote monitoring for blood glucose readings, sending them to other people that are features that some people utilise. Yeah, I I wonder within that are there situation or yeah, examples of times or situations that you wouldn't want that data shared with any of those groups particularly?

P07

I I wouldn't say this is me personally, but I know Dexcom you can obviously do the sharing with um anybody but feel like if you've got somebody that's quite overbearing, you possibly don't want them to see your data all the time because they'll just constantly ask if you're OK and that would get. It's nice that they're concerned, but it would get a bit annoying if they're every time something happened they were asking what was going on. And I know for me, like my mum and dad are on it because I'm not at home for uni. So, I think for them it gives them a bit more peace of mind and I don't mind them seeing it but I kind of see in certain cases where you wouldn't want people to see your data.

Interviewer

Yeah, does anyone else have any views on that? What about you, [P16]? Is there? How do you feel about kind of sharing that data and are there times you would rather or times you would rather not?

P16

So when I was away at uni, my parents could also see my levels and everything, which I didn't mind because I think it was useful, especially at night time. Um. And now since being on the 780 it comes with an app. So, I can see my level on my phone and everything, and then I can give it to whoever and they can see it as well. So I have, so my mum and dad have it and also my fiancé has it too and and actually I don't mind them seeing that because it helps me to feel a bit more relaxed, especially if I'm not near any of them. Or like so I'm now doing a Masters course part time, so I'll be travelling into uni on my own and I think that will help me feel more relaxed knowing that other people can see my level and everything, especially as I'll be driving and stuff.

Interviewer

Yeah, definitely. That makes a lot of sense. Yeah. Um. I guess going forward, kind of, there's the idea that there's. I mean, as part of what this study's looking into as well is including even more data from wearable devices and that kind of compounding and building. I just wondered how people felt about that and if they have any opinions on that, particularly with the link to then privacy of this growing pool of data?

P07

I think with the watch there's maybe a little more concern about privacy because it can track like the location of your workout. So obviously we don't want people to know where you live and where you do your daily activities, so as long as that information is removed then I would feel OK with it. And now with Apple Watch, I already share data with like friends and family and do the competition. So, I don't mind that so much, but it's definitely the location I wouldn't want people seeing my location.

Interviewer

Yeah, that makes a lot of sense that how what do other people think on that similar or any differences?

P02

Yeah. Agree with with [P07]. You just don't want anyone to like, be able to track you down. Well, that would be slightly concerning. Yeah, I feel like integrating data. Again, if it makes if it improves people's lives, improves opportunities with technology like you know. I don't think like this kind of data is like any more than. I don't think it faces that different problems from any other kind of data that we have problems with now, like patient data generally. It's just everything's at risk of being stolen in, you know, dark web or whatever. It's just, I think, the same precautions just need to be applied all round or better precautions.

Interviewer

No. Great. I think as a final point I just. One thing I'd be interested here is just kind of again kind of reviewing that my watch use over the last month compared to kind of the other two or three months as part of the study. Has anyone noticed any, any significant changes in their usage of it or points they think are interesting more general?

P07

I think I've been using it around the same amount as I have before. The only times I've not used it is if I've forgot my charger and I've gone home or something or I just didn't charge it in the morning and it's died. But I would say I've been trying to use it about the same amount I've I'm basically always wearing it unless it's charging.

Interviewer

Yeah, yeah. I guess it's kind of it was said at the beginning um yeah [P07], [P02], [P19] you're all using um Apple Watches you were using before. Yeah. Yeah, maybe [P16] as you were kind of testing out one of the Fitbits, have you seen that kind of usage change in the last month, maybe compared to the previous months?

P16

I think I've been using it about the same. So after the, not the last focus group, but the one before that, I then started to wear overnight and I've just continued to do that. So, the only time I'm not wearing it is when it's charging. Um. So I think for the last few months it's been my usage has just being the same but more than it was like at the start.

Interviewer

Right. I see. And during that time, has anything kind of, have you seen any points that you think “ohh this would be”. Any points that either you found quite useful part of your type one diabetes management or anything that's kind of triggered thoughts of “ohh if I had this, this would be really helpful”.

P16

Not particularly. Nothing really stands out.

Interviewer

Yeah, what about anyone else? Has anyone kind of had an yeah anything come to them in the last month or so of usage?

P07

I think just what I said earlier about linking the activity type that would be useful um for me, that's probably the main thing I can think of just now.

Interviewer

Yeah. And I think that is a, yeah, a really interesting point and particularly with I guess the yeah, the Apple Watch where you're inputting that information, could be easily transferred across.